A-A-54115
21 March 1990

COMMERCIAL ITEM DESCRIPTION

CEMENT, ZINC OXIDE AND EUGENOL, DENTAL

The General Services Administration has authorized the use of this Commerical Item Description as a replacement for Military Specification MIL-C-36548A, which is cancelled.

This commercial item description covers zinc oxide and eugenol dental cement suitable for use in the final cementation of precision-fitting crowns and fixed partial dentures.

Salient characteristics:

The cement package shall consist of a minimum of 30.0 grams (g) of cement powder, a minimum of 14.0 milliliters (ml) of liquid, a plastic dropper, a plastic powder scoop, and a set of instructions for use.

Powder. The composition of the cement powder shall be essentially zinc oxide, iron oxide pigment, a zinc acetate accelerator, and high molecular weight polymethyl methacrylate. Cement powder shall contain no abrasive fillers such as quartz or alumina. The quality of the materials shall be as follows:

a. Zinc oxide shall be in accordance with the U.S.P. for lead and arsenic contents only.

b. Iron oxide pigment shall be of drug or cosmetic grade.

c. Zinc acetate shall be U.S.P.

d. The polymethyl methacrylate shall be nontoxic. The manufacturer shall submit a statement to the contracting officer that the polymer furnished has no toxic effect on the normal, healthy person.

Liquid. The composition of the liquid shall be eugenol, U.S.P. with acetic acid, A.C.S. added as an accelerator.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Accessories. The powder scoop shall be fabricated of molded polystyrene and shall be of 1/4-teaspoon capacity.

Instructions. Adequate and accurate instructions for using the dental cement shall accompany each unit package. In addition, the instructions shall include setting time, mixing time, powder-liquid ratio, and methods for proportioning and manipulating the cement, including number of drops of liquid required to mix with one scoop of powder to make a mix of 2.5 to 3.0 g to 1 ml (powder-liquid) ratio. Two copies of instructions shall be forwarded to the contracting officer for information purposes.

The cement shall not form lumps, granules, or evolve gas when mixed according to manufacturer's instructions. The color of the set cement shall be light yellow.

Physical properties. The cement shall have the following physical properties when tested as specified below. The setting time shall be 8.5 to 12.0 minutes. The compressive strength shall be 7600 pounds per square inch, minimum. The solubility and disintegration shall be 0.3 percent by weight, maximum. The film thickness shall be 33 microns (0.0013 inch), maximum. The consistency shall be 22 millimeters, minimum.

Immediate container. Immediate container for the cement powder shall be a plastic bottle. Immediate container for the liquid shall be an amber glass bottle with screw cap closure. Each bottle shall be permanently and legibly marked with the lot number and date of manufacture. Each bottle shall be further marked with the following legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

Delivery schedule. Not more than 4 months shall have elapsed from the date of manufacture of the cement to the date of delivery to the Government.

Workmanship. The powder and liquid shall be free from all dirt, foreign materials, and impurities which detract from their appearance or which may impair their serviceability.

Unit. Package (PG). One package containing one bottle of powder, one bottle of liquid, a dropper, a powder scoop, and instructions for use, as specified, constitutes one unit. All components shall be packaged in a suitable box.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the used of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of cement. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1. AQL 1.0.

Examination. The dental cement shall be examined for defects including, but not limited to, those listed below:

Defects.

Package does not contain all specified components.
Powder not homogeneous or free of extraneous material.
Liquid not clear or has significant color.
Bottle for liquid not amber or does not have screw cap closure.
Loose caps, broken bottles, leaks, etc. present.

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Marking of immediate containers not as specified. Instructions missing, incorrect, incomplete, or illegible.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Preparation of test specimens. The preparation of test specimens shall be conducted at a temperature of 70 to 75^{0}F, and 45 to 50 percent relative humidity with a powder-liquid ratio of 2.5 g to 1.0 ml (standard consistency).

Consistency. Procedure for consistency test shall be as described in American Dental Association (ADA) Specification No. 30 for Dental Zinc Oxide-Eugenol Type Restorative Materials, except that the 120 gram load shall be applied for 7 minutes.

Setting time. A mix of cement using 1.25 g of powder and 0.5 ml of liquid shall be loaded into a polyethylene crown form cap with approximate internal dimensions of 9.5 mm diameter by 7 mm deep. Type A durometer hardness shall be determined as described in ASTM designation D2240. Readings shall be taken at 0.5 minute intervals. Hardness shall be plotted against time, and the setting time in minutes shall be the interpolated value of time when durometer reading equals 50. Setting time shall be reported to the nearest 0.5 minute.

Compressive strength. Procedure for compressive strength test shall be as described in ADA Specification No. 30, except that the specimens shall be stored for one week in water at 37^{0}C before crushing. Test equipment used for determining compressive strength shall be the Instron Corporation's Test Machine, or other suitable equipment, provided prior approval is received from the contracting officer. Testing conditions on the Instron test machine shall be as follows:

a. Crosshead speed of 0.2 inch per minute.
b. Force range of 1000 pounds.
c. Chart speed of 10 inches per minute.

The value for compressive strength shall be reported as the average of three or more from a group of five specimens and shall be rounded off to the nearest 50 pounds. If the values for individual specimens fall more than 15 percent below the average of the five, they shall be discarded and the average of the remaining specimens shall be reported. In case more than two of the specimens are eliminated, the test shall be repeated.

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Solubility and disintegration. Procedure for solubility and disintegration test shall be as described in ADA Specification No. 30.

Film thickness. Procedure for film thickness test shall be as described in ADA Specification No. 30.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and of regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-935-4121	CEMENT, ZINC OXIDE AND EUGENOL, DENTAL, 30 Gram

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

48650

PROJECT NO. 6520-2338
LOCATION: NCR - DISK 8

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